

I. STEP-BY-STEP NUMERICAL DRY RUN ON URBANSOUND8K:

To demonstrate the exact algebraic mechanics of **Algorithm 2 (Phase 3: Entropy-Aware Choquet Meta-Learner)** in a complex acoustic environment, we trace an edge-case sample from the **UrbanSound8K (US8K)** benchmark ($K = 10$ classes, $\ln(K) = \ln(10) \approx 2.3026$).

Acoustic Context: The test clip is a distant, reverberant recording of urban **Drilling** (**Ground Truth** $y^* = 5$, Class **Drilling**) captured near a congested street intersection:

- **Local Branch (B_L — CNN14) Failure:** The sharp, high-frequency percussive impact shocks over-activate the localized 7×1 temporal auxiliary kernels, causing B_L to falsely identify the clip as the heavier percussive impulses of **Class 8 (Jackhammer)**.
- **Global Branch (B_G — HTS-AT) Failure:** The continuous, low-frequency electric motor hum and traffic drone deceive the global Swin patch-attention mechanism, causing B_G to misclassify the clip as **Class 6 (Engine Idling)**.
- **Linear Ensemble Failure:** Arithmetic averaging ($\frac{p_L + p_G}{2}$) succumbs to the loudest single-branch distractor and predicts **Class 8 (Jackhammer)**.
- **ECHO-Net Outcome:** The Choquet Meta-Learner dynamically attenuates discordant single-branch outliers via entropy gating, pools concordant evidence along the joint capacity pedestal for **Class 5 (Drilling)**, and leverages the residual MLP to achieve a decisive, correct classification.

A. Step 1: Raw Logits and Temperature Scaling

As specified in Algorithm 2 (Lines 4–5), the frozen backbones output raw continuous logit vectors $z_L, z_G \in \mathbb{R}^{10}$. Using the learned class-wise temperature vectors $T_L, T_G \in \mathbb{R}^{10}$, the softened probability distributions $p_i = \text{Softmax}\left(\frac{z_i}{\text{Softplus}(T_i) + \epsilon}\right)$ for $i \in \{L, G\}$ are computed as:

$$z_L = [0.76, 0.48, 0.76, 0.98, 2.66, 1.16, 0.48, \mathbf{3.27}, 0.98, 0.98]^T \quad (1)$$

$$z_G = [0.98, 0.48, 0.76, 0.76, 2.56, \mathbf{3.15}, 0.48, 1.16, 1.48, 0.98]^T \quad (2)$$

$$\text{Softplus}(T_L) = [0.9741, 0.9741, 0.9741, 0.9741, 1.0701, 0.9741, 0.9741, 1.0375, 0.9741, 0.9741]^T \quad (3)$$

$$\text{Softplus}(T_G) = [0.9741, 0.9741, 0.9741, 0.9741, 1.0701, 1.0375, 0.9741, 0.9741, 0.9741, 0.9741]^T \quad (4)$$

Evaluating the Softmax yields the base probabilities p_L and p_G :

$$p_L = [0.0400, 0.0300, 0.0400, 0.0502, 0.2203, 0.0603, 0.0300, \mathbf{0.4288}, 0.0502, 0.0502]^T \quad (5)$$

$$\Rightarrow \hat{y}_L = \arg \max(p_L) = \mathbf{8} \quad (\text{Jackhammer} - \mathbf{FAIL})$$

$$p_G = [0.0519, 0.0310, 0.0414, 0.0414, 0.2074, \mathbf{0.3949}, 0.0310, 0.0624, 0.0867, 0.0519]^T \quad (6)$$

$$\Rightarrow \hat{y}_G = \arg \max(p_G) = \mathbf{6} \quad (\text{Engine Idling} - \mathbf{FAIL})$$

Failure of Conventional Static Averaging:

$$p_{\text{mean}} = \frac{p_L + p_G}{2} = [0.0459, 0.0305, 0.0407, 0.0458, 0.2139, 0.2276, 0.0305, \mathbf{0.2456}, 0.0684, 0.0510]^T \quad (7)$$

The arithmetic mean predicts $\hat{y}_{\text{mean}} = \arg \max(p_{\text{mean}}) = \mathbf{8}$ (Jackhammer), completely failing to recover Ground Truth Class $y^* = 5$ (Drilling)!

B. Step 2: Shannon Entropy Dynamic Confidence Gating

In Algorithm 2 (Line 6), the scalar confidence scores $C_L, C_G \in [0, 1]$ are calculated from the normalized Shannon entropy across all $K = 10$ classes:

$$H(p_L) = - \sum_{k=1}^{10} p_L[k] \ln(p_L[k] + \epsilon) = 1.7842 \Rightarrow C_L = 1 - \frac{1.7842}{\ln(10)} = \mathbf{0.2251} \quad (8)$$

$$H(p_G) = - \sum_{k=1}^{10} p_G[k] \ln(p_G[k] + \epsilon) = 1.8643 \Rightarrow C_G = 1 - \frac{1.8643}{\ln(10)} = \mathbf{0.1903} \quad (9)$$

Because probability mass is dispersed across secondary candidate classes, both models exhibit elevated entropy, driving confidence scores down to $C_L \approx 0.23$ and $C_G \approx 0.19$, which dynamically restrains their singleton capacities.

C. Step 3: Axiomatic Fuzzy Capacities

Following Algorithm 2 (Lines 8–9), the gated singleton capacities $\mathbf{g}_L, \mathbf{g}_G$ and the bounded joint capacity $\mathbf{g}_{\text{joint}}$ are computed from learnable vectors $\gamma_L, \gamma_G, \gamma_{\text{joint}} \in \mathbb{R}^{10}$:

$$\begin{aligned} \mathbf{g}_L &= C_L \cdot \sigma(\gamma_L) = 0.2251 \cdot \sigma(\gamma_L) \\ &= [0.1182, 0.1182, 0.1182, 0.1265, \mathbf{0.1623}, 0.1265, 0.1182, 0.1322, 0.1265, 0.1265]^T \end{aligned} \quad (10)$$

$$\begin{aligned} \mathbf{g}_G &= C_G \cdot \sigma(\gamma_G) = 0.1903 \cdot \sigma(\gamma_G) \\ &= [0.0999, 0.0999, 0.0999, 0.0999, \mathbf{0.1353}, 0.1140, 0.0999, 0.1022, 0.1140, 0.1070]^T \end{aligned} \quad (11)$$

The monotonic joint capacity $\mathbf{g}_{\text{joint}} = \max(\mathbf{g}_L, \mathbf{g}_G) + \sigma(\gamma_{\text{joint}}) \odot (1 - \max(\mathbf{g}_L, \mathbf{g}_G))$ guarantees that $\mathbf{g}_{\text{joint}} \geq \max(\mathbf{g}_L, \mathbf{g}_G)$:

$$\mathbf{g}_{\text{joint}} = [0.6501, 0.6501, 0.6501, 0.6651, \mathbf{0.8498}, 0.6940, 0.6501, 0.6979, 0.6858, 0.6755]^T \quad (12)$$

D. Step 4: Vectorized 2-Source Choquet Integration

Following Algorithm 2 (Lines 11–15), the unnormalized Choquet integral is computed element-wise:

$$p_{\text{choquet}}[k] = \begin{cases} (\mathbf{p}_L[k] - \mathbf{p}_G[k]) \cdot \mathbf{g}_L[k] + \mathbf{p}_G[k] \cdot \mathbf{g}_{\text{joint}}[k], & \text{if } \mathbf{p}_L[k] \geq \mathbf{p}_G[k] \\ (\mathbf{p}_G[k] - \mathbf{p}_L[k]) \cdot \mathbf{g}_G[k] + \mathbf{p}_L[k] \cdot \mathbf{g}_{\text{joint}}[k], & \text{if } \mathbf{p}_G[k] > \mathbf{p}_L[k] \end{cases} \quad (13)$$

Critical Contrast on Contending Classes:

- 1) **Class 5 (Drilling, True Class $y^* = 5$, $\mathbf{p}_L[5] = 0.2203 \geq \mathbf{p}_G[5] = 0.2074$):**

$$\begin{aligned} p_{\text{choquet}}[5] &= (0.2203 - 0.2074) \cdot 0.1623 + 0.2074 \cdot 0.8498 \\ &= (0.0129) \cdot 0.1623 + 0.1763 = 0.0021 + 0.1763 = \mathbf{0.1784} \end{aligned} \quad (14)$$

Mechanism: Both branches concur that Class 5 holds significant acoustic evidence ($\min \approx 0.21$). Multiplied by the dominant cooperative capacity $\mathbf{g}_{\text{joint}}[5] = 0.8498$, this shared foundation yields a massive unnormalized value of **0.1784**.

- 2) **Class 8 (Jackhammer, Local Distractor, $\mathbf{p}_L[8] = 0.4288 \geq \mathbf{p}_G[8] = 0.0624$):**

$$\begin{aligned} p_{\text{choquet}}[8] &= (0.4288 - 0.0624) \cdot 0.1322 + 0.0624 \cdot 0.6979 \\ &= (0.3664) \cdot 0.1322 + 0.0435 = 0.0484 + 0.0435 = \mathbf{0.0919} \end{aligned} \quad (15)$$

Mechanism: Although B_L expresses high confidence (0.4288), B_G heavily refutes it (0.0624). The single-branch excess (0.3664) is crushed by the small gated singleton capacity (0.1322).

- 3) **Class 6 (Engine Idling, Global Distractor, $\mathbf{p}_G[6] = 0.3949 > \mathbf{p}_L[6] = 0.0603$):**

$$\begin{aligned} p_{\text{choquet}}[6] &= (0.3949 - 0.0603) \cdot 0.1140 + 0.0603 \cdot 0.6940 \\ &= (0.3346) \cdot 0.1140 + 0.0418 = 0.0381 + 0.0418 = \mathbf{0.0799} \end{aligned} \quad (16)$$

Summing across all 10 classes ($\sum_{k=1}^{10} p_{\text{choquet}}[k] = 0.5442$) and normalizing (Line 16) yields:

$$\mathbf{p}_{\text{choquet}} = [0.0500, 0.0361, 0.0481, 0.0526, \mathbf{0.3278}, 0.1470, 0.0361, 0.1690, 0.0709, 0.0626]^T \quad (17)$$

The Choquet Integral alone reverses the erroneous ranking, elevating Ground Truth Class 5 (Drilling) to 32.78%, while Jackhammer falls to 16.90% and Engine Idling to 14.70%!

E. Step 5: Residual Correction MLP and Final Bounded Fusion

As specified in Algorithm 2 (Lines 18–24), the concatenated logit vector $\mathbf{z}_{\text{concat}} = [\mathbf{z}_L \parallel \mathbf{z}_G] \in \mathbb{R}^{20}$ passes through the Residual MLP $\mathcal{R}$. Detecting the dual-branch acoustic confusion pattern, $\mathcal{R}$ generates correction logits $\mathbf{z}_{\text{res}}$, yielding the correction probability vector $\Delta \mathbf{p} = \text{Softmax}(\mathbf{z}_{\text{res}})$:

$$\mathbf{z}_{\text{res}} = [-0.20, -0.40, -0.20, -0.10, \mathbf{1.80}, -0.80, -0.50, -0.80, -0.20, -0.30]^T \quad (18)$$

$$\Delta \mathbf{p} = [0.0664, 0.0544, 0.0664, 0.0734, \mathbf{0.4908}, 0.0365, 0.0492, 0.0365, 0.0664, 0.0601]^T \quad (19)$$

With the learned scalar parameter $\alpha_{\text{res}} = 0.50 \implies \sigma(\alpha_{\text{res}}) = 0.6225$, the final bounded fused probability distribution $\mathbf{p}_{\text{fused}}$ is:

$$\begin{aligned} \mathbf{p}_{\text{fused}}^{\text{unnorm}} &= \mathbf{p}_{\text{choquet}} + \sigma(\alpha_{\text{res}}) \cdot \Delta \mathbf{p} \\ &= [0.0500 + 0.0413, 0.0361 + 0.0339, \dots, \mathbf{0.3278} + \mathbf{0.3055}, \dots]^T \\ &= [0.0913, 0.0700, 0.0894, 0.0983, \mathbf{0.6333}, 0.1697, 0.0667, 0.1917, 0.1122, 0.1000]^T \end{aligned} \quad (20)$$

Normalizing $\mathbf{p}_{\text{fused}}$ to sum strictly to 1.0 (Line 24):

$$\mathbf{p}_{\text{fused}} = [0.0563, 0.0431, 0.0551, 0.0606, \mathbf{0.3903}, 0.1046, 0.0411, 0.1181, 0.0692, 0.0616]^T \quad (21)$$

$$\hat{y}_{\text{ECHO-Net}} = \arg \max(\mathbf{p}_{\text{fused}}) = \mathbf{5} \quad (\text{Drilling} \implies \mathbf{DECISIVE SUCCESS!}) \quad (22)$$

F. Comprehensive 10-Class Numerical Summary

TABLE I
COMPLETE 10-CLASS PROBABILITY TRAJECTORY ACROSS ALL STAGES OF ECHO-NET ON URBANSOUND8K

Class k	UrbanSound8K Class Name	Local $B_L (p_L)$	Global $B_G (p_G)$	Static Mean	Capacity g_{joint}	Choquet (p_{choquet})	Residual Δp	ECHO-Net (p_{fused})
1	Air Conditioner	0.0400	0.0519	0.0459	0.6501	0.0500	0.0664	0.0563
2	Car Horn	0.0300	0.0310	0.0305	0.6501	0.0361	0.0544	0.0431
3	Children Playing	0.0400	0.0414	0.0407	0.6501	0.0481	0.0664	0.0551
4	Dog Bark	0.0502	0.0414	0.0458	0.6651	0.0526	0.0734	0.0606
5	Drilling ($y^* = 5$)	0.2203	0.2074	0.2139	0.8498	0.3278	0.4908	0.3903 (Correct)
6	Engine Idling	0.0603	0.3949	0.2276	0.6940	0.1470	0.0365	0.1046
7	Gun Shot	0.0300	0.0310	0.0305	0.6501	0.0361	0.0492	0.0411
8	Jackhammer	0.4288	0.0624	0.2456	0.6979	0.1690	0.0365	0.1181
9	Siren	0.0502	0.0867	0.0684	0.6858	0.0709	0.0664	0.0692
10	Street Music	0.0502	0.0519	0.0510	0.6755	0.0626	0.0601	0.0616
Predicted Class (arg max)		8 (FAIL)	6 (FAIL)	8 (FAIL)	—	5 (SUCCESS)	—	5 (DECISIVE)